Critical Path For N4 Version 2 Solid Rocket Booster

Nakuja Rocketry Club – Solid Propulsion Team
Critical Path Documentation: N4 Version 2 Solid Rocket Booster
Design → Static Test (3-Week Cycle)

Target: Complete one full 4-grain SRB design-to-static-test cycle within **3 weeks**. After a successful static test, produce **two flight-ready SRBs** (one per week) for early-September dual launch.

Overall Goal: Reliable 4-grain KNSB BATES motor delivering ~2.5 kN peak thrust for 3.5 km apogee.

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1. Critical Path Overview

The critical path is the sequence of dependent tasks that determines the shortest possible time to complete the cycle. Any delay on these tasks delays the entire project.

Critical Path Sequence:

1. Design (OpenMotor + OpenRocket) →
2. Purchasing →
3. Parallel Fabrication (3D printing + Machining + Grain cooking) →
4. Grain curing (1 week) + concurrent non-grain tasks →
5. Dry assembly → Wet assembly →
6. Static fire test

Total Target Duration: 21 days (3 weeks)

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2. Team Roles & Responsibilities

Sub-team	Members	Primary Responsibilities
Propellant Team	2	Liner fabrication, grain cooking & curing, ignitor preparation, chemistry inventory
Machining Team	2	Design & machine nozzle (Mild Steel), bulkhead (Al 7075), core rods (Aluminium); casing preparation
Ignition Team	2	Ignition circuit design/maintenance, trigger system, loadcell setup & maintenance, new ignitor housing
3D Printing Team	1	Design & print: casting tools (top/bottom plates), ignitor & trigger circuit housing
Documentation Team	1	Daily logs, grain criteria table, procedure records, static test report, GitHub updates, photo documentation

Cross-team expectation: Every member must understand the full process and support other sub-teams when needed. Documentation is mandatory for all.

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3. Detailed 3-Week Timeline (Critical Path)

Week 1: Design + Purchasing + Early Fabrication
Days 1–3 (Design)

- Requirements: 3.5 km apogee, 2.5 kN peak thrust
- Run OpenMotor simulations for 4-grain BATES geometry
- Validate with OpenRocket trajectory & trajectory
- Freeze final dimensions (core diameter, grain length, nozzle throat, bulkhead)
- **Owner:** Machining Lead + Propellant Lead (joint review)
- **Deliverable:** Dimension sheet + simulation report

Days 2–4 (Purchasing – Parallel)

- Sorbitol
- Al 7075 round rods (bulkhead) - 106 mm OD
- Mild Steel (nozzle) - 106 mm OD

- Aluminium core rods - 50 mm OD
- PLA filament - High Speed
- Crafting paper, epoxy, brushes/rollers
- Springs, nuts, compression hardware
- Ziplock bags
- 1 pyrogenic ignitor
- **Owner:** Designated purchasing member (coordinate bulk order) - BenCollins Mureithi, Alex
- **Deliverable:** All materials received and logged

Days 3–7 (Early Fabrication)

- 3D Printing: 4 pairs of casting tools (top + bottom plates) – 1 day
- Machining: Core rods (multiple diameters) – 2 days per rod (parallel where possible)
- Fabricate liners (crafting paper + epoxy)
- **Owner:** 3D Printing (1 day), Machining Team (ongoing)

Week 2: Main Fabrication + Curing Start
Days 8–14

- **Propellant Team:**
  - Cook 4 grains (2 per day)
  - Start 7-day cure under compression
- **Machining Team:**
  - Machine nozzle (3 days)
  - Machine bulkhead (3 days) – run in parallel
- **Ignition Team:**
  - Design/print new ignitor housing
  - Prepare/test ignition circuit & pressure sensor
- **3D Printing:** Support any remaining tools
- **Documentation:** Update grain criteria table daily

Critical Note: Grain curing occupies 7 days. Use this time for test-stand reinforcement, load-cell calibration, and dry-assembly preparation.

Week 3: Assembly + Static Test
Days 15–17 (Dry Assembly)

- Bolt bulkhead + nozzle to stainless-steel casing
- Verify alignment and sealing
- **Owner:** Machining + Propellant (3 days)

Days 18 (Wet Assembly)

- Insert 4 cured grains
- Install ignitor
- Final motor inspection
- **Owner:** Full team (1 day)

Days 19–21 (Static Test Preparation & Fire)

- Mount on test stand
- Connect load cell + data system
- Safety briefing + dry run
- Static fire (4-grain motor)
- Record thrust curve + video
- Immediate post-test inspection
- **Owner:** Entire Solid Propulsion Team
- **Deliverable:** Static test report + data package

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4. Priority Order & Dependencies

Priority	Task	Depends On	Owner	Duration
1	Design (OpenMotor/OpenRocket)	—	Joint	2–3 days
2	Purchasing	Design	Purchasing lead	2–3 days
3	3D Casting tools	Purchasing	3D Printing	1 day

Priority	Task	Depends On	Owner	Duration
4	Core rods machining	Purchasing	Machining	2 days each
5	Liner + Grain cooking	Casting tools + core rods	Propellant	2 days cooking
6	Nozzle + Bulkhead	Purchasing	Machining	3 days each (parallel)
7	Grain curing	Cooking	Propellant	7 days
8	Dry assembly	Nozzle + Bulkhead	Machining	3 days
9	Wet assembly	Cured grains	Propellant + Machining	1 day
10	Static test	Full assembly	Full team	1–2 days

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5. Post-Static-Test Plan (Two Flight Motors)

After successful static test:

- Week 4: Re-use SRB #1 (repeat critical path at accelerated pace)
- Week 5: Produce SRB #2
- Target: Both motors ready for dual launch in early September

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6. Daily Operating Rules

- Working hours: 9:00 AM – 5:00 PM (lunch 1–2 PM)
- Daily photo + activity summary required
- Clean workstation by 4:45 PM
- All changes and data logged in GitHub by Documentation Team

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7. Risk Mitigation

Risk	Mitigation
Delayed materials	Consolidate purchasing early; maintain buffer stock
Grain voids/cracks	Pressure curing + vacuum if available; microscope checks
Machining delays	Parallel nozzle/bulkhead work; prioritize critical items
Curing delay	Start cooking as soon as tools are ready
Test stand issues	Use curing week for reinforcement and load-cell calibration